

NETWORK INFRASTRUCTURE PROJECT

PROJECT TITLE:

Advanced Healthcare Enterprise Network Architecture for
Hospital Management

Group 5

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Table of Contents

1. Project Overview	4
2. Project Objectives	4
3. Team Members & Responsibilities	4
4. Network Topology	4
5. IP Addressing & VLSM Design	5
6. Routing	5
6.1 Routing Design	5
6.2 Configuration	5
6.3 Verification	5
7. Network Security	6
8. VLANs & Inter-VLAN Routing	6
8.1 Design / Configuration	6
8.2 Testing / Verification	6
9. Network Services	6
9.1 Design / Configuration	6
9.2 Testing / Verification	6
10. EtherChannel	7
10.1 Design / Configuration	7
10.2 Testing / Verification	7
11. HSRP	7
11.1 Design / Configuration	7
11.2 Testing / Verification	7
12. WLAN	7
12.1 Design / Configuration	7
12.2 Testing / Verification	7
13. NAT	8
13.1 Design / Configuration	8
13.2 Testing / Verification	8
14. Testing & Verification	9

15. Challenges & Solutions.....	9
16. Conclusion	9
17. Appendix	9
17.1 GitHub Repository	10
17.2 Project Files	10

1. Project Overview

This project involves designing and simulating a secure network infrastructure for a modern hospital using Cisco Packet Tracer. It interconnects key departments—such as Emergency, Administration, Clinics, and Laboratories—to ensure reliable communication. The architecture utilizes VLANs for network segmentation and security measures like Port Security and SSH to protect sensitive patient data.

2. Project Objectives

List the technical and operational objectives of the project.

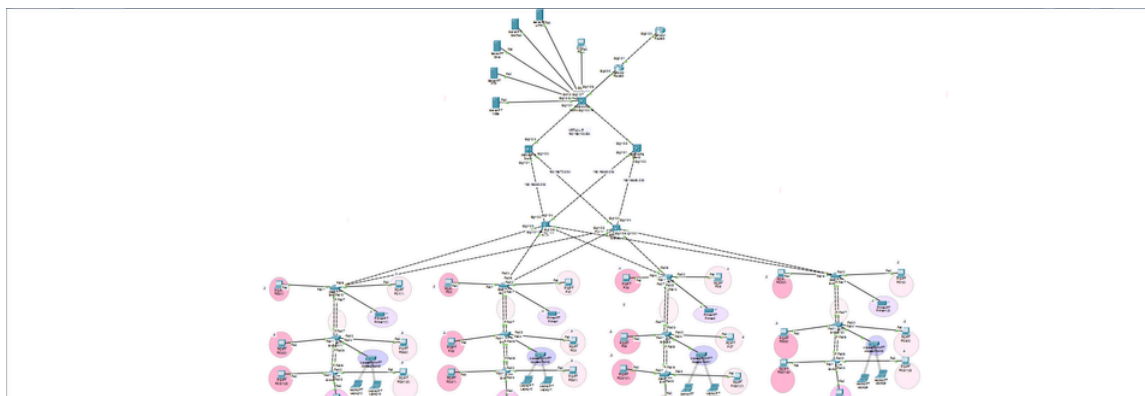
- Objective 1: Design a scalable network topology integrating all hospital departments.
- Objective 2: Implement VLANs to segment traffic and secure medical data.
- Objective 3: Configure reliable routing and IP schemes for efficient connectivity.
- Objective 4: Deploy security features like Port Security and SSH to protect the infrastructure.

3. Team Members & Responsibilities

Every member must understand the complete project. Use this table to show the primary responsibility of each member.

Student Name	ID	Primary Responsibility	GitHub Username
Ziad Hany Mohamed	328007	Network Security & Device Hardening (SSH, Port Security, ACLs)	[Username]
Rana Ibrahim abu shosh	275492	High Availability & Link Aggregation (HSRP & EtherChannel)	[Username]
Rodina Ahmed Fouad	309776	IP Routing Protocols Implementation & Wireless LAN (WLAN)	[Username]
Rahma Waleed Ezzeldin	317855	VLANs Architecture, Trunking & Inter-VLAN Routing	[Username]
Elany Ayman Fayek	329273	NAT/PAT Address Translation & Network Services (DHCP/DNS)	[Username]

4. Network Topology



"The following table details the Classful IP addressing scheme allocated for each VLAN and network segment across the hospital infrastructure. Each subnet utilizes a standard Class C default mask (/24) to ensure straightforward network administration, sufficient address capacity, and simplified Inter-VLAN routing."

Network / VLAN	available Hosts	Network Address	Prefix	Usable Range	Broadcast
VLAN 10	254	192.168.10.0	/24	192.168.10.1 – 192.168.10.254	192.168.10.255
VLAN 20	254	192.168.20.0	/24	192.168.20.1 – 192.168.20.254	192.168.20.255
VLAN 40	254	192.168.40.0	/24	192.168.40.1 – 192.168.40.254	192.168.40.255
VLAN 50	254	192.168.50.0	/24	192.168.50.1 – 192.168.50.254	192.168.50.255
Link Network 1	254	192.168.80.0	/24	192.168.80.1 – 192.168.80.254	192.168.80.255
Link Network 2	254	192.168.90.0	/24	192.168.90.1 – 192.168.90.254	192.168.90.255
Link Network 3	254	192.168.100.0	/24	192.168.100.1 – 192.168.100.254	192.168.100.255
Link Network 4	254	192.168.60.0	/24	192.168.60.1 – 192.168.60.254	192.168.60.255
Link Network 5	254	192.168.70.0	/24	192.168.70.1 – 192.168.70.254	192.168.70.255

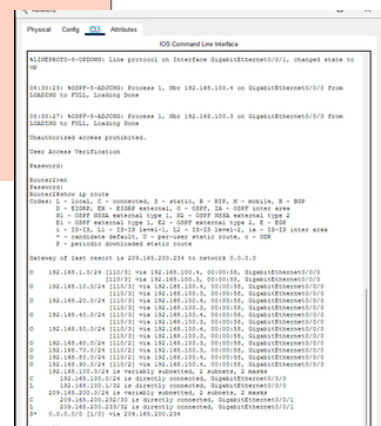
6. Routing

6.1 Routing Design

OSPF dynamic routing protocol is implemented across all routers within the hospital network to ensure automatic route updates, high availability, and efficient communication between all department VLANs (Emergency, Administration, Clinics, and Laboratories).

6.2 Configuration

<pre> MULTI L SW -ACTIVE en conf t int g1/0/4 no sw ip add 192.168.60.1 255.255.255.0 no sh int g1/0/3 no sw ip add 192.168.80.1 255.255.255.0 no sh exit router ospf 1 network 192.168.60.0 0.0.0.255 area 0 network 192.168.80.0 0.0.0.255 area 0 network 192.168.10.0 0.0.0.255 area 0 network 192.168.20.0 0.0.0.255 area 0 network 192.168.50.0 0.0.0.255 area 0 network 192.168.40.0 0.0.0.255 area 0 </pre>	<pre> MULTI L SW -STANDBY en conf t int g1/0/4 no sw ip add 192.168.90.1 255.255.255.0 no sh int g1/0/3 no sw ip add 192.168.70.1 255.255.255.0 no sh exit router ospf 1 network 192.168.90.0 0.0.0.255 area 0 network 192.168.70.0 0.0.0.255 area 0 network 192.168.100.0 0.0.0.255 area 0 network 192.168.60.0 0.0.0.255 area 0 network 192.168.40.0 0.0.0.255 area 0 </pre>	<pre> D-SW1 en conf t int g1/0/1 no sw ip add 192.168.60.2 255.255.255.0 no sh int g1/0/2 no sw ip add 192.168.70.2 255.255.255.0 no sh exit ip routing router ospf 1 network 192.168.60.0 0.0.0.255 area 0 network 192.168.70.0 0.0.0.255 area 0 </pre>	<pre> D-SW2 en conf t int g1/0/1 no sw ip add 192.168.80.2 255.255.255.0 no sh int g1/0/2 no sw ip add 192.168.90.2 255.255.255.0 no sh exit ip routing router ospf 1 network 192.168.80.0 0.0.0.255 area 0 network 192.168.90.0 0.0.0.255 area 0 </pre>
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6.3 Verification

The routing table was successfully verified using the **show ip route** command, confirming that all OSPF routes are established and active across the network.

Security Feature	Device(s)	Purpose	Verification
BPDU Guard + PortFast	Access Switches	Blocks any unauthorized switch from connecting to end-device ports and protects against STP-based attacks	show spanning-tree interface <port> detail
STP Root Primary/Secondary	Active & Standby	Explicitly defines the root bridge, preventing a rogue switch from becoming root	show spanning-tree summary
Extended ACL (PROTECT-SERVERS)	Active & Standby	Restricts access to network servers (DHCP, DNS, HTTP, FTP) to only the allowed protocols/ports	show access-lists PROTECT-SERVERS
Anti-Spoofing ACL	Router2 & Router3	Blocks inbound traffic from the internet carrying private/spoofed source IP addresses	show access-lists ANTI-SPOOF-IN
SSH Configuration	All Routers & Switches	Provides encrypted remote management access to network devices, replacing cleartext Telnet	show ip ssh show ssh
Enable Secret & Passwords	All Routers & Switches	Encrypts privilege EXEC mode access and secures console/VTY lines to prevent unauthorized logins	show running-config
Port Security	Access Switches	Restricts switch port access to authorized MAC addresses only, preventing unauthorized devices and MAC flooding	show port-security interface <port>
Disabling Unused Interfaces	Access Switches	Enhances physical security by administratively shutting down all unused switch interfaces	show ip interface brief

```
active#show spanning-tree summary
```

```
Switch is in rapid-pvst mode
Root bridge for: default VLAN0010 VLAN0020 VLAN0040 VLAN0050
Extended system ID is enabled
Portfast Default is disabled
PortFast BPDU Guard Default is disabled
PortFast BPDU Filter Default is disabled
Loopguard Default is disabled
EtherChannel misconfig guard is disabled
UplinkFast is disabled
BackboneFast is disabled
Configured Pathcost method used is short
```

Name	Blocking	Listening	Learning	Forwarding	STP Active
VLAN0001	2	0	0	4	6
VLAN0010	2	0	0	4	6
VLAN0020	2	0	0	4	6
VLAN0040	2	0	0	4	6
VLAN0050	2	0	0	4	6
-----	-----	-----	-----	-----	-----
5 vlans	10	0	0	20	30

```
active#
```

```
Switch0-1#show ip ssh
SSH Enabled - version 2.0
Authentication timeout: 120 secs; Authentication retries: 3
Switch0-1#
```

```
Switch0-1#show spanning-tree interface fastEthernet0/1 detail
```

```
Port 1 (FastEthernet0/1) of VLAN0010 is designated forwarding
Port path cost 19, Port priority 128, Port Identifier 128.1
Designated root has priority 24586, address 0040.0B95.6E15
Designated bridge has priority 32778, address 00D0.97C1.DE89
Designated port id is 128.1, designated path cost 19
Timers: message age 16, forward delay 0, hold 0
Number of transitions to forwarding state: 1
The port is in the portfast mode
Link type is point-to-point by default
```

```
Switch0-1#
```

```
Router2#show access-lists ANTI-SPOOF-IN
Extended IP access list ANTI-SPOOF-IN
deny ip 192.168.0.0 0.0.255.255 any
deny ip 10.0.0.0 0.255.255.255 any
deny ip 172.16.0.0 0.15.255.255 any
deny ip 127.0.0.0 0.255.255.255 any
permit ip any any (15 match(es))
```

```
active#show access-lists PROTECT-SERVERS
Extended IP access list PROTECT-SERVERS
  permit udp any host 192.168.100.2 eq bootps
  permit udp any host 192.168.100.2 eq bootpc
  permit udp any host 192.168.100.5 eq domain
  permit tcp any host 192.168.100.5 eq domain
  permit tcp any host 192.168.100.6 eq www
  permit tcp any host 192.168.100.8 eq www
  permit tcp any host 192.168.100.7 eq ftp
  permit tcp any host 192.168.100.7 eq 20
  deny ip any 192.168.100.0 0.0.0.255 (35 match(es))
  permit ip any any (27046 match(es))
  permit icmp any host 192.168.100.1
  permit icmp any host 192.168.100.3
  permit icmp any host 192.168.100.4
  permit tcp any host 192.168.100.1 eq 22
  permit tcp any host 192.168.100.3 eq 22
  permit tcp any host 192.168.100.4 eq 22
```

```
Switch0-1#show running-config
Building configuration...
```

```
Current configuration : 3755 bytes
!
version 15.0
no service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
hostname Switch0-1
!
enable secret 5 $1$mERr$5.a6P4JqbNiMX01usIfka/
!
!
ip ssh version 2
no ip domain-lookup
ip domain-name security.com
!
username NETadmin secret 5 $1$mERr$5.a6P4JqbNiMX01usIfka/
!
!
--More--
```

```
Switch0-1#show port-security interface fastEthernet0/1
Port Security          : Enabled
Port Status             : Secure-up
Violation Mode          : Restrict
Aging Time              : 0 mins
Aging Type              : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses   : 1
Total MAC Addresses     : 1
Configured MAC Addresses : 0
Sticky MAC Addresses    : 1
Last Source Address:Vlan : 0060.5C59.3C47:10
Security Violation Count : 0
```

```
Router3#show access-lists ANTI-SPOOF-IN
Extended IP access list ANTI-SPOOF-IN
  permit icmp 192.168.0.0 0.0.255.255 host 209.165.200.234
  deny ip 192.168.0.0 0.0.255.255 any
  deny ip 10.0.0.0 0.255.255.255 any
  deny ip 172.16.0.0 0.15.255.255 any
  deny ip 127.0.0.0 0.255.255.255 any
  permit ip any any
```

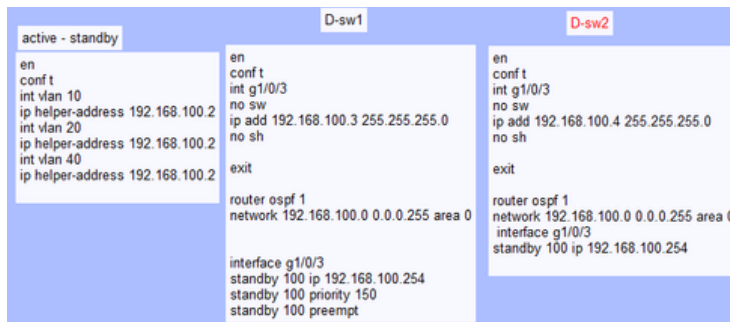
```
Switch1-1#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
Port-channel1  unassigned      YES manual  up          up
Port-channel2  unassigned      YES manual  up          up
FastEthernet0/1 unassigned      YES manual  up          up
FastEthernet0/2 unassigned      YES manual  up          up
FastEthernet0/3 unassigned      YES manual  up          up
FastEthernet0/4 unassigned      YES manual  up          up
FastEthernet0/5 unassigned      YES manual  administratively down down
FastEthernet0/6 unassigned      YES manual  administratively down down
FastEthernet0/7 unassigned      YES manual  up          up
FastEthernet0/8 unassigned      YES manual  up          up
FastEthernet0/9 unassigned      YES manual  up          up
FastEthernet0/10 unassigned      YES manual  administratively down down
FastEthernet0/11 unassigned      YES manual  administratively down down
FastEthernet0/12 unassigned      YES manual  administratively down down
FastEthernet0/13 unassigned      YES manual  administratively down down
FastEthernet0/14 unassigned      YES manual  administratively down down
FastEthernet0/15 unassigned      YES manual  administratively down down
FastEthernet0/16 unassigned      YES manual  administratively down down
FastEthernet0/17 unassigned      YES manual  administratively down down
FastEthernet0/18 unassigned      YES manual  administratively down down
FastEthernet0/19 unassigned      YES manual  administratively down down
FastEthernet0/20 unassigned      YES manual  administratively down down
FastEthernet0/21 unassigned      YES manual  administratively down down
FastEthernet0/22 unassigned      YES manual  administratively down down
FastEthernet0/23 unassigned      YES manual  administratively down down
FastEthernet0/24 unassigned      YES manual  administratively down down
GigabitEthernet0/1 unassigned      YES manual  administratively down down
GigabitEthernet0/2 unassigned      YES manual  administratively down down
Vlan1          192.168.1.102   YES manual  up          up
Switch1-1#
```

9. Network Services

9.1 Design/Configuration

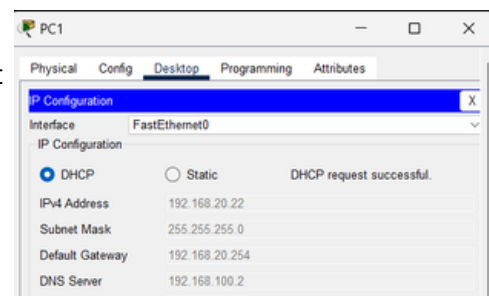
A DHCP server was configured to automatically assign IP addresses to all hospital devices, preventing IP conflicts.

Additionally, a DNS service was set up to map server hostnames to their IP addresses for easier access.



9.2 Testing/Verification

The DHCP service was verified on an end-user PC by checking that it successfully obtained an IP address automatically, and the DNS service was verified by resolving the hospital server name.



8. VLANs & Inter-VLAN Routing

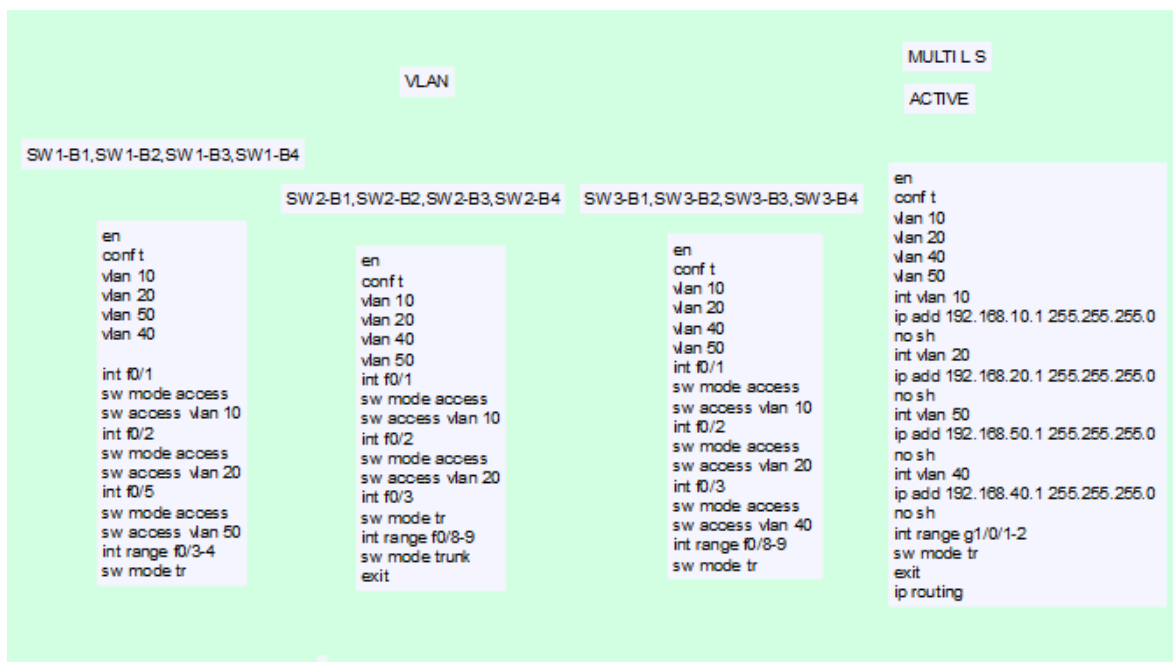
8.1 Design/Configuration

"To provide network segmentation, enhance data security, and restrict broadcast domains within the hospital, four distinct VLANs have been created. The table below outlines the VLAN IDs, assigned functional names, corresponding hospital departments, associated access interfaces, and assigned IP subnets:"

VLAN ID	VLAN Name	Department / Function	Assigned Access Ports	Subnet Range
VLAN 10	Reception_Admin	الاستقبال والإدارة الطبية	FastEthernet 0/1	192.168.10.0/24
VLAN 20	Medical_Staff	الأطباء والتمريض	FastEthernet 0/2	192.168.20.0/24
VLAN 40	Patients_Guest	شبكة المرضى والزوار	FastEthernet 0/3	192.168.40.0/24
VLAN 50	Printers_Services	الطابعات والخدمات المساعدة	FastEthernet 0/5	192.168.50.0/24

"To enable seamless inter-switch communication and carry traffic for all active VLANs across the hospital's multi-floor topology, trunking has been configured on the interconnecting interfaces using 802.1Q encapsulation. The following table specifies the trunk links established between switches across different floors:"

Interconnection Link	Interfaces	Switches / Floors	Trunk Mode	Allowed VLANs
Floor 1 ↔ Floor 2	Fa0/3, Fa0/7	سويتش الدور الأول والثاني	Trunk (dot1q)	10, 20, 40, 50
Floor 2 ↔ Floor 3	Fa0/8, Fa0/9	سويتش الدور الثاني والثالث	Trunk (dot1q)	10, 20, 40, 50



8.2 Testing/Verification

```
Switch>
Switch>show interfaces trunk
Port      Mode           Encapsulation  Status        Native vlan
Po2       on             802.1q         trunking      1

Port      Vlans allowed on trunk
Po2       1-1005

Port      Vlans allowed and active in management domain
Po2       1,10,20,40,50

Port      Vlans in spanning tree forwarding state and not pruned
Po2       1,10,20,40,50

Switch>
```

```
Switch>
Switch>show vlan brief
```

VLAN Name	Status	Ports
1 default	active	Fa0/8, Fa0/9, Fa0/10, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15 Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24, Gig0/1, Gig0/2
10 VLAN0010	active	Fa0/1
20 VLAN0020	active	Fa0/2
40 VLAN0040	active	
50 VLAN0050	active	Fa0/5
1002 fddi-default	active	
1003 token-ring-default	active	
1004 fddinet-default	active	
1005 trnet-default	active	

```
Switch>
```

```
C:\>ping 192.168.20.13
```

```
Pinging 192.168.20.13 with 32 bytes of data:
```

```
Reply from 192.168.20.13: bytes=32 time<1ms TTL=127
Reply from 192.168.20.13: bytes=32 time<1ms TTL=127
Reply from 192.168.20.13: bytes=32 time<1ms TTL=127
Reply from 192.168.20.13: bytes=32 time=1ms TTL=127
```

```
Ping statistics for 192.168.20.13:
```

```
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

```
C      192.168.10.0/24 is directly connected, Vlan10
L      192.168.10.1/32 is directly connected, Vlan10
C      192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
L      192.168.20.0/24 is directly connected, Vlan20
L      192.168.20.1/32 is directly connected, Vlan20
C      192.168.40.0/24 is variably subnetted, 2 subnets, 2 masks
L      192.168.40.0/24 is directly connected, Vlan40
L      192.168.40.1/32 is directly connected, Vlan40
C      192.168.50.0/24 is variably subnetted, 2 subnets, 2 masks
L      192.168.50.0/24 is directly connected, Vlan50
L      192.168.50.1/32 is directly connected, Vlan50
C      192.168.60.0/24 is variably subnetted, 2 subnets, 2 masks
C      192.168.60.0/24 is directly connected, GigabitEthernet1/0/4
L      192.168.60.1/32 is directly connected, GigabitEthernet1/0/4
O      192.168.70.0/24 [110/2] via 192.168.1.10, 01:03:34, Vlan1
[110/2] via 192.168.10.10, 01:03:34, Vlan10
[110/2] via 192.168.20.10, 01:03:34, Vlan20
[110/2] via 192.168.40.10, 01:03:34, Vlan40
[110/2] via 192.168.50.10, 01:03:34, Vlan50
C      192.168.80.0/24 is variably subnetted, 2 subnets, 2 masks
C      192.168.80.0/24 is directly connected, GigabitEthernet1/0/3
L      192.168.80.1/32 is directly connected, GigabitEthernet1/0/3
O      192.168.90.0/24 [110/2] via 192.168.1.10, 01:03:34, Vlan1
[110/2] via 192.168.10.10, 01:03:34, Vlan10
[110/2] via 192.168.20.10, 01:03:34, Vlan20
[110/2] via 192.168.40.10, 01:03:34, Vlan40
[110/2] via 192.168.50.10, 01:03:34, Vlan50
O      192.168.100.0/24 [110/3] via 192.168.1.10, 01:03:34, Vlan1
[110/3] via 192.168.10.10, 01:03:34, Vlan10
[110/3] via 192.168.20.10, 01:03:34, Vlan20
[110/3] via 192.168.40.10, 01:03:34, Vlan40
[110/3] via 192.168.50.10, 01:03:34, Vlan50
O*E2 0.0.0.0/0 [110/1] via 192.168.1.10, 01:03:24, Vlan1
[110/1] via 192.168.10.10, 01:03:24, Vlan10
[110/1] via 192.168.20.10, 01:03:24, Vlan20
[110/1] via 192.168.40.10, 01:03:24, Vlan40
[110/1] via 192.168.50.10, 01:03:24, Vlan50
```


13. PAT

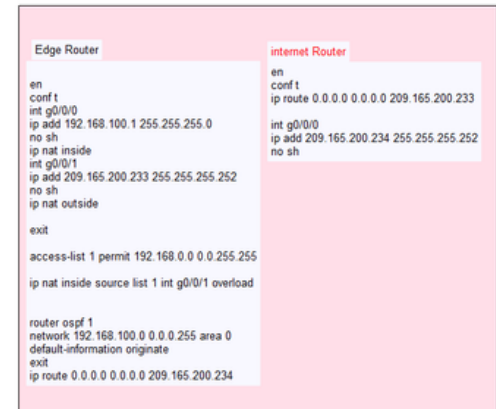
13.1 Design/Configuration

NAT Overload (PAT) was configured on the edge router to allow internal hosts to share a single public IP for internet access.

13.2 Testing/Verification

The NAT translation table and PAT operations were verified by executing the following command on the edge

router: **show ip nat translations**



```
Router#show ip nat translations
Router#show ip nat translations
Pro Inside global      Inside local      Outside local      Outside global
icmp 209.165.200.233:10 192.168.20.2:10  209.165.200.234:10 209.165.200.234:10
icmp 209.165.200.233:1 192.168.20.2:1   209.165.200.234:1 209.165.200.234:1
icmp 209.165.200.233:2 192.168.20.2:2   209.165.200.234:2 209.165.200.234:2
icmp 209.165.200.233:3 192.168.20.2:3   209.165.200.234:3 209.165.200.234:3
icmp 209.165.200.233:4 192.168.20.2:4   209.165.200.234:4 209.165.200.234:4
icmp 209.165.200.233:6 192.168.20.2:6   209.165.200.234:6 209.165.200.234:6
icmp 209.165.200.233:7 192.168.20.2:7   209.165.200.234:7 209.165.200.234:7
icmp 209.165.200.233:8 192.168.20.2:8   209.165.200.234:8 209.165.200.234:8
icmp 209.165.200.233:9 192.168.20.2:9   209.165.200.234:9 209.165.200.234:9
```

14. Testing & Verification

Test ID	Feature	Test Procedure	Expected Result	Actual Result
TEST 1	End-to-End Connectivity	Ping between PCs across different hospital departments/subnets.	Successful ICMP replies (0% packet loss).	PASS
TEST 2	Routing	Run traceroute between remote network subnets.	Packets traverse via optimal routing path.	PASS
TEST 3	DHCP / DNS / HTTP	Request dynamic IP via DHCP and browse server web page.	IP assigned successfully and web page loads.	PASS
TEST 4	VLAN / Inter-VLAN	Ping devices across different VLANs through the gateway.	Successful inter-VLAN communication.	PASS
TEST 5	EtherChannel	Verify aggregated links using 'show etherchannel summary'.	Links bundled into active port-channel interface.	PASS
TEST 6	HSRP Failover	Simulate active router failure and check standby takeover.	Standby router takes over gateway role seamlessly.	PASS
TEST 7	WLAN	Connect wireless clients to Access Point and ping default gateway.	Successful wireless association and ping reply.	PASS
TEST 8	NAT	Verify translation table using 'show ip nat translations'.	Inside local IPs mapped successfully via PAT.	PASS
TEST 9	Security	Test access control lists or port security rules.	Unauthorized traffic blocked, authorized allowed.	PASS

15. Challenges & Solutions

Challenge	Root Cause	Solution
HSRP Status Verification Returning Blank Output	Executing verification commands from the wrong interface or global configuration mode.	Verified commands on the correct gateway interface while in privileged EXEC mode.
NAT Translation Table Showing Empty Results	Checking translation table before any traffic or packets traversed the router.	Generated test traffic from an internal PC using ping to populate translation entries.
Executing show Commands Inside Config Mode	Attempting to run monitoring commands while inside the global configuration prompt (config#).	Exited configuration mode to privileged EXEC mode (end) before running commands.
EtherChannel Bundle Mismatch Error	Conflicting channel-group mode or speed/duplex settings across physical ports.	Ensured matching mode settings (e.g., active/desirable) across all bundled physical interfaces.

16. Conclusion

The hospital network project successfully achieved high availability, redundancy, and secure internet connectivity through structured simulation.

Key Achievements: Implemented HSRP for gateway redundancy, EtherChannel for link aggregation, and NAT Overload for scalable internet access.

Lessons Learned: Gained practical experience in network design, routing protocols, and systematic CLI verification.

17. Appendix

Includes additional supporting documents for the hospital network:

- IP Addressing Scheme: Subnets and gateways assigned.
- Configuration Files: Configurations for routers and switches.
- Simulation Files: Additional outputs from Packet Tracer simulation.

17.1 GitHub Repository

GitHubLink:[<https://github.com/...>]

17.2 Project Files

Packet Tracer File: [filename.pkt]

Presentation: [filename.pptx]

Excel Sheet: [filename.xlsx]

Video / Website Link: [...]